



บริษัท ทีโอเอ-โดฟเคมี อินดัสตรีส์ จำกัด
TOA-DOVECHEM INDUSTRIES CO., LTD.



: ISO 9001
: ISO 14001
: ISO 50001
: OHSAS 18001

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SAFETY DATA SHEET
OF
FORMALDEHYDE (FORMALIN)
FM-40%





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SAFETY DATA SHEET

Formaldehyde (Formalin) FM-40%

1 IDENTIFICATION OF THE SUBSTANCE/ MIXTURE AND OF THE SUPPLIER

1.1 Product identifiers

Product name : Formaldehyde (FORMALIN) FM-40%

Molecular Formula : CH₂O

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses : Manufacture of substances such as Urea Formaldehyde resin, Melamine formaldehyde resin, Phenolic resin

1.3 Details of the supplier of the safety data sheet

Company : TOA-DOVECHEM INDUSTRIES CO.,LTD.

31/5 Moo3 Bangna-Trad Rd.,KM.23 Bangsaothong, Samutprakarn 10540
Thailand

Telephone : +66 27400822-26

2 HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 [EU-GHS/CLP]

Acute toxicity, Oral (Category 4)

Acute toxicity, Inhalation (Category 3)

Acute toxicity, Dermal (Category 3)

Skin corrosion (Category 2)

Skin sensitization (Category 1)

Carcinogenicity (Category 1)

Eye damage, (Category 2A)

Respiratory irritation, (Category 1)

Specific target organ toxicity, (Category 1)



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2.2 Label elements and Precautionary statement(s)

GHS Label:



Hazard statement(s)

- Toxic if swallowed
- Toxic if contact with skin
- Causes severe skin burns and eye damage
- May cause an allergic skin reaction.
- Toxic if inhaled.
- May cause respiratory irritation.
- Suspected of causing cancer.
- Causes damage to organ

Precautionary statement(s)

- Do not breathe dust/ fume/ gas/ mist/ vapours/ spray.
- Wear protective gloves/ protective clothing/ eye protection/ face protection.
- IF SWALLOWED : Immediately call a POISON CENTER or doctor/physician.
- IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
- Immediately call a POISON CENTER or doctor/ physician.

Risk phrases

- R23/ 24/ 25 Toxic by inhalation, in contact with skin and if swallowed.
R34 Causes burns.
R40 Limited evidence of a carcinogenic effect.

Safety-phrases

- S1/2 Keep containers tightly closed.
S26 In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
S36/ 37/ 39 Wear suitable protective clothing, gloves and eye/face protection.
S45 In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible).
S51 Use only in well-ventilated areas.





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3 COMPOSITION/ INFORMATION ON INGREDIENTS

3.1 Mixtures

Chemical name and concentration : Formaldehyde (FORMALIN) FM-40%

Synonyms : Formalin, Methanol, Methylene Oxide, Oxymethylene

Formula : CH₂O

Molecular Weight : 30,03 g/mol

3.2 Component Classification Concentration

Composition	CAS-No..	Ingredient (%)	Un Number	EC (EINECS)
3.2.1 Formaldehyde	50-00-00	40	2209	200-001-8
3.2.2 Methanol	67-56-1	9		
3.2.3 Water		51		

4 FIRST AID MEASURES

4.1 Description of first aid measures

General advice

Consult a physician. Show this safety data sheet to the doctor in attendance.

If inhaled

If breathed in, move person into fresh air. If not breathing, give artificial respiration. Consult a physician.

In case of skin contact

Wash plenty of water. Take victim immediately to hospital. Consult a physician.

In case of eye contact

Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.

If swallowed

Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Rinse mouth with water. Consult a physician.

4.2 Most important symptoms and effects, both acute and delayed

Warning: contains methanol. May be fatal or cause blindness if swallowed. Cannot be made nonpoisonous.





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5 FIREFIGHTING MEASURES

5.1 Extinguishing media

Suitable extinguishing media

Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

5.2 Special hazards arising from the substance or mixture

Carbon oxides

5.3 Advice for firefighters

Wear self contained breathing apparatus for fire fighting if necessary.

5.4 Further information

Use water spray to cool unopened containers.



6 ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Wear respiratory protection. Avoid breathing vapors, mist or gas. Ensure adequate ventilation.

Remove all sources of ignition. Evacuate personnel to safe areas. Beware of vapours accumulating to form explosive concentrations. Vapours can accumulate in low areas.

6.2 Environmental precautions

Prevent further leakage or spillage if safe to do so. Do not let product enter drains. Discharge into the environment must be avoided.

6.3 Methods and materials for containment and cleaning up

Contain spillage, and then collect with an electrically protected vacuum cleaner or by wet-brushing and place in container for disposal according to local regulations (see section 13). Keep in suitable, closed containers for disposal.

6.4 Reference to other sections

For disposal see section 13.

7 HANDLING AND STORAGE

7.1 Precautions for safe handling

Avoid contact with skin and eyes. Avoid inhalation of vapour or mist.

Keep away from sources of ignition - No smoking. Take measures to prevent the build up of electrostatic charge.



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7.2 Conditions for safe storage, including any incompatibilities

Store in cool place. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage.

8 EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Components with workplace control parameters

IDLH	: 20 ppm (NIOSH, 2005)
PEL-STEL	: 2 ppm (OSHA, 2006)
PEL-TWA	: 0.75 ppm (OSHA, 2006)
TLV-STEL	: 10 ppm
TLV-TWA	: 3 ppm
TLV-C	: 5 ppm



8.2 Exposure controls

Appropriate engineering controls

Avoid contact with skin, eyes and clothing. Wash hands before breaks and immediately after handling the product.

Personal protective equipment

Eye/face protection

Tightly fitting safety goggles. Use equipment for eye protection tested and approved under appropriate government standards

Skin protection

Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique to avoid skin contact with this product.

Body Protection

Complete suit protecting against chemicals, The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

Respiratory protection

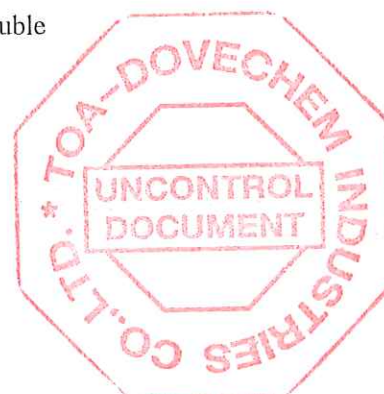
Where risk assessment shows air-purifying respirators are appropriate use a full-face respirator with multi-purpose combination. If the respirator is the sole means of protection, use a full-face supplied air respirator or self contained breathing apparatus; SCBA



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9 PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: Form : liquid, clear	Colour: colourless
Odour	: pungent	
Boiling point	: 99 °C	
Vapour pressure	: 4.2 mmHg , 0.55 Kfa	
Density	: 1.098	
Flash point	: 75-85 °C	
Upper/lower flammability or explosive limits	: Upper explosion limit : 73% Lower explosion limit : 7%	
Melting point	: -118 °C	
Water solubility	: completely soluble	
Evaporation rate	: -	
pH	: 2.8 - 4.0	



10 STABILITY AND REACTIVITY

10.1 Chemical stability

Contains the following stabilizer(s) : Methanol

10.2 Conditions to avoid

Heat, flames and sparks. Heating under high pressure in closed container

10.3 Incompatible materials

Incompatible materials	Reaction
Amine	Exothermic reaction
AZO Compounds	Exothermic reaction
Caustics	Heat generation and violent polymerization
Dithiocarbamates	Formation of flammable gases and toxic fumes, formation of CS ₂
Alkali & alkali earth metals	Heat generation and formation of hydrogen flammable gas
Nitrides	Heat generation,, formation of ammonia gas and violent polymerization
Nitro compound	Heat generation
Unsaturated aliphatics and sulfides	Heat generation
Organic peroxides	Violent reaction
Oxidizing agents	Heat generation, fire and decomposition
Reducing agents	Heat generation, formation of flammable gases



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10.4 Hazardous decomposition products

Carbonmonoxide, Carbondioxide, Hydrogen, toxic fume

11 TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

LD50 : 10 mg/kg

Respiratory or skin sensitization : May cause allergic skin reaction.

Potential health effects

Inhalation Toxic if inhaled. Causes respiratory tract irritation.

Ingestion Toxic if swallowed.

Skin Toxic if absorbed through skin. Causes skin irritation.

Eyes Causes eye burns.

Signs and Symptoms of Exposure

Warning: contains methanol. May be fatal or cause blindness if swallowed. Cannot be made nonpoisonous.

12 ECOLOGICAL INFORMATION

12.1 Bioaccumulative potential

no data available

12.2 Mobility in soil

no data available

12.3 Other adverse effects

Harmful to aquatic life.

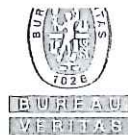


13 DISPOSAL CONSIDERATIONS

: Any disposal practice must be in compliance with local, state or federal laws and regulations.

(Contact local or environmental agency for specific rules.)

Do not dump into sewers, on the ground or into any water.



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14 TRANSPORT INFORMATION

14.1 Trade name : FORMALDEHYDE SOLUTION with not less than 25 % formaldehyde

14.1 UN number : 2209

14.3 Fixed Tank : L4BN

14.4 Packaging group : III

15 REGULATORY INFORMATION

This safety datasheet complies with the requirements of Regulation (EC) No. 1907/2006.

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Hazardous Substance Act B.E. 2535

15.2 Chemical Safety Assessment

no data available

16 OTHER INFORMATION

Reference : Manual handling dangerous chemicals. (Formaldehyde) / Department of Industrial Works

Prepared by : TOA-Dovechem Industries Co., Ltd.

